AI33-MPOPT · THIRD PILLAR · 2026

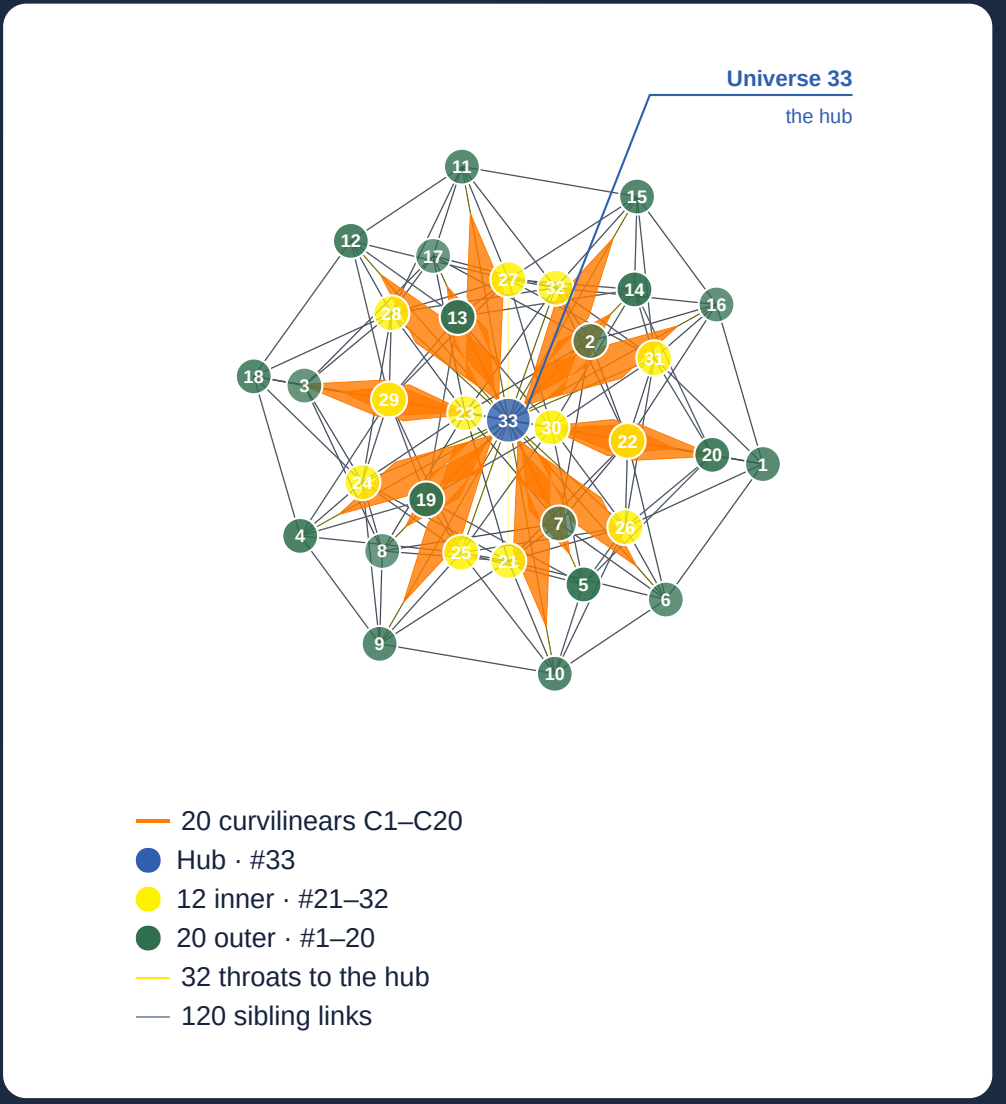
Loop Quantum Gravity Bridge

The 33-node discrete operator model set beside loop quantum gravity: what corresponds, what it computes, and its seen and unseen halves.

THE ARCHITECTURE AND ITS COLOUR CODE

1	12	20	20
hub · #33	inner · #21–32	outer · #1–20	curvilinear

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A third pillar — with its scope stated first

AI33-MPOPT now has three research lines: a string/M-theory echo (the 11-mode band and its invariant 3-forms), a quantum-gravity bridge (the information–curvature law), and this file — a comparison with loop quantum gravity.

LQG and AI33-MPOPT share one genuine philosophical starting point: geometry is built from a discrete combinatorial structure, with no background spacetime assumed. That shared starting point is real and worth examining.

Each page is marked by kind: context, analogy, computed, program or open.

WHERE THIS FILE STANDS

AI33-MPOPT builds geometry from a discrete 33-node structure, with its own operator and its own information–curvature law.

This file sets its computed results beside LQG's structures: spin networks, area and volume spectra, holonomies, spin foams and dimensional flow.

LQG has its own propagators, amplitudes and computed spectra.

Pages 10–12 add the loop-quantum computations: spin on all 33 nodes, the 720° sign sequence and the $\sin^2(\omega t)$ density.

Loop quantum gravity in one page

Spin networks

Graphs whose edges carry SU(2) representation labels. Edge labels give quanta of area; nodes give quanta of volume.

Quantized geometry

Area and volume operators have discrete spectra — this is LQG's central achievement, derived not assumed.

Holonomies

Curvature enters through parallel transport around closed loops of the connection — the loop in 'loop quantum gravity'.

Hamiltonian constraint

Dynamics must satisfy a constraint algebra. This is genuinely hard and remains actively debated.

Spin foams

Histories of spin networks; these give transition amplitudes — LQG's propagator formalism.

Dimensional flow

Several approaches (LQG, CDT, asymptotic safety) predict spectral dimension running from ≈4 at large scales to ≈2 at small.

ANALOGY

What genuinely corresponds

Background-free start	Geometry from a graph, no metric assumed	*Same philosophical starting point — real*
Discrete quanta	33 nodes, 152 edges, fixed combinatorics	*Spin ½ on the 120 surface links and spin 1 on the 32 throats close all 33 nodes (pages 10–12)*
Operator-driven	T = L + pK + (1−p)Q, self-adjoint, positive	*Analogy to an evolution operator — NOT a constraint*
Geometry from spectra	Eigenvalues/eigenvectors carry the structure	*Shared with LQG, Connes, spectral geometry generally*
Discrete curvature	Transport curvature R from Wasserstein cost	*A different discrete curvature than holonomy; the closed loop returns −1 after one turn and +1 after two*

COMPUTED

What it does

THE OPERATOR

$T = L + pK + (1-p)Q$ is self-adjoint and positive. Its Green function $G = T^{-1}$ exists once the 32 throats couple the hub to the shell.

THE LAW

$L\,R = \kappa(F - \langle F \rangle)$, exact at first order on the icosahedron. Coupling 0.987 ± 0.003, top of 17 graph families and the most stable; 13 of 17 families carry signal (criterion set before testing).

THE LOOP

The Rivero-Zeta family is a closed loop: $t = 0$ and $t = 1$ are the same object. Carried once around it, the oriented relations return negated; carried twice, unchanged.

THE ANGLES

The icosahedron's own angles run through all 33: 63.4349° between neighbouring apexes, 41.8103° between neighbouring face centres, 72° five-fold at an apex, 120° three-fold at a face. 33 steps of 10.9091° and 32 of 11.25° each close to 360°.

THE ORIENTED HALF

Of 23 admissible relations, reversal leaves 17 unchanged and negates 6. T, G and I(i,j) are symmetric, so they read the same on both sides; the 6 oriented relations tell the sides apart.

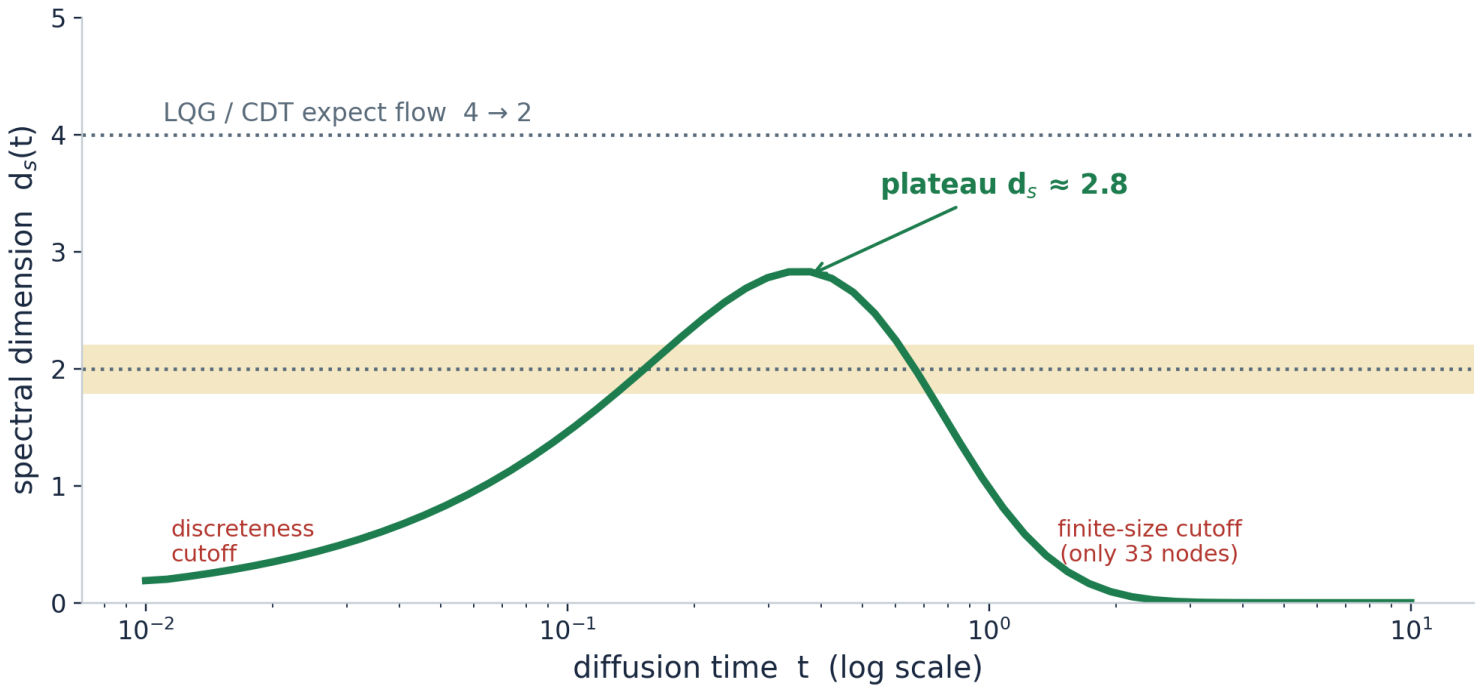
THE UNSEEN HALF

Six dimensions: three seen, three unseen, and the unseen three are a complete second copy. The hub is where the observer stands, and its 32 throat cones cover the whole sky at once.

Sources: this file, pages 6–9; The Rivero-Zeta Curve, pages 5, 11 and 13–14; The 33-Multiverse Map, pages 16–18.

COMPUTED

Spectral dimension on the 33-node graph



Spectral dimension $d_S(t) = -2 \cdot d \ln P(t) / d \ln t$, computed directly from the graph Laplacian's heat kernel.

THE RESULT

d_S plateaus near 2.8 (33-node graph) and converges to 2.56 at 642 nodes. Measured on the subdivided graph: $d_S \approx 2.56$. The 4 → 2 flow is the LQG comparison point.

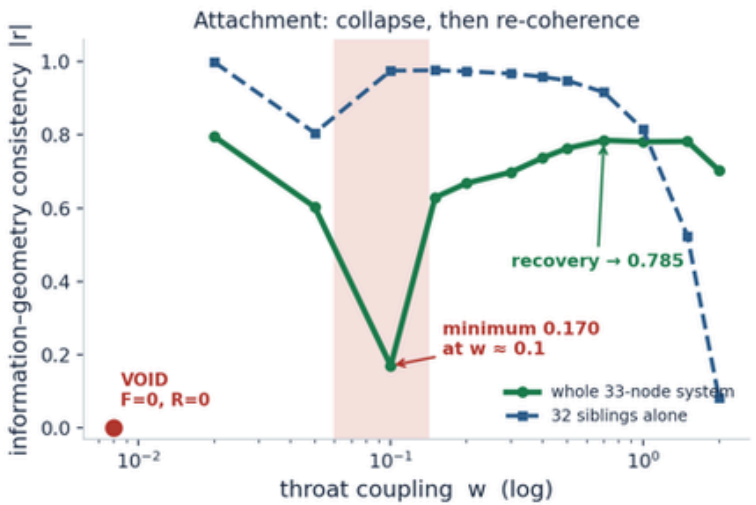
TESTED BY GROWING THE GRAPH

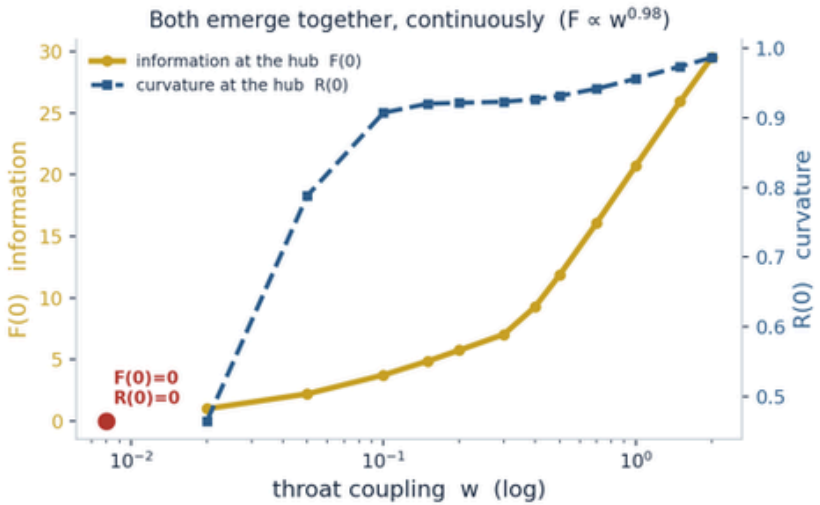
Geodesic subdivision to 42, 162 and 642 nodes gives d_S → 2.56 and stops. This is not a finite-size artifact: the model sits near 2.5, a triangulated 2-sphere, and no scale produces 4 → 2.

Heat kernel averaged over all 33 nodes at once.

COMPUTED

The Void — lawfulness without relations





WHAT EXISTS IN THE VOID

Set the throat coupling to zero and the operator T becomes SINGULAR — it has no inverse, so there is no Green function and no propagator.

Every quantity vanishes exactly:
$F(0) = 0 \quad R(0) = 0$
$G_{00} = 0 \quad \max|G_{0j}| = 0$

No information. No curvature. No influence arriving. Not assumed — computed.

NO PROPAGATOR → NO 'HOW LONG'

Distance, travel time and causal order are all defined through the propagator. With no relations there is no propagator, so questions of duration have no answer in the void — not unknown, undefined.

THE SIBLINGS ARE ALREADY LAWFUL

While the hub sits at zero, the 32-sibling structure alone holds the law at 0.988. Lawfulness does not begin at inception; the hub enters a structure that is already coherent.

ATTACHMENT: COLLAPSE, THEN RE-COHERENCE

Weak coupling is disruptive: whole-system consistency falls to 0.170 at w ≈ 0.1. Above w ≈ 0.15 the structure re-coheres, rising to 0.785 by w ≈ 0.7. Inception has a threshold.

QUANTUM AND GRAVITY EMERGE TOGETHER

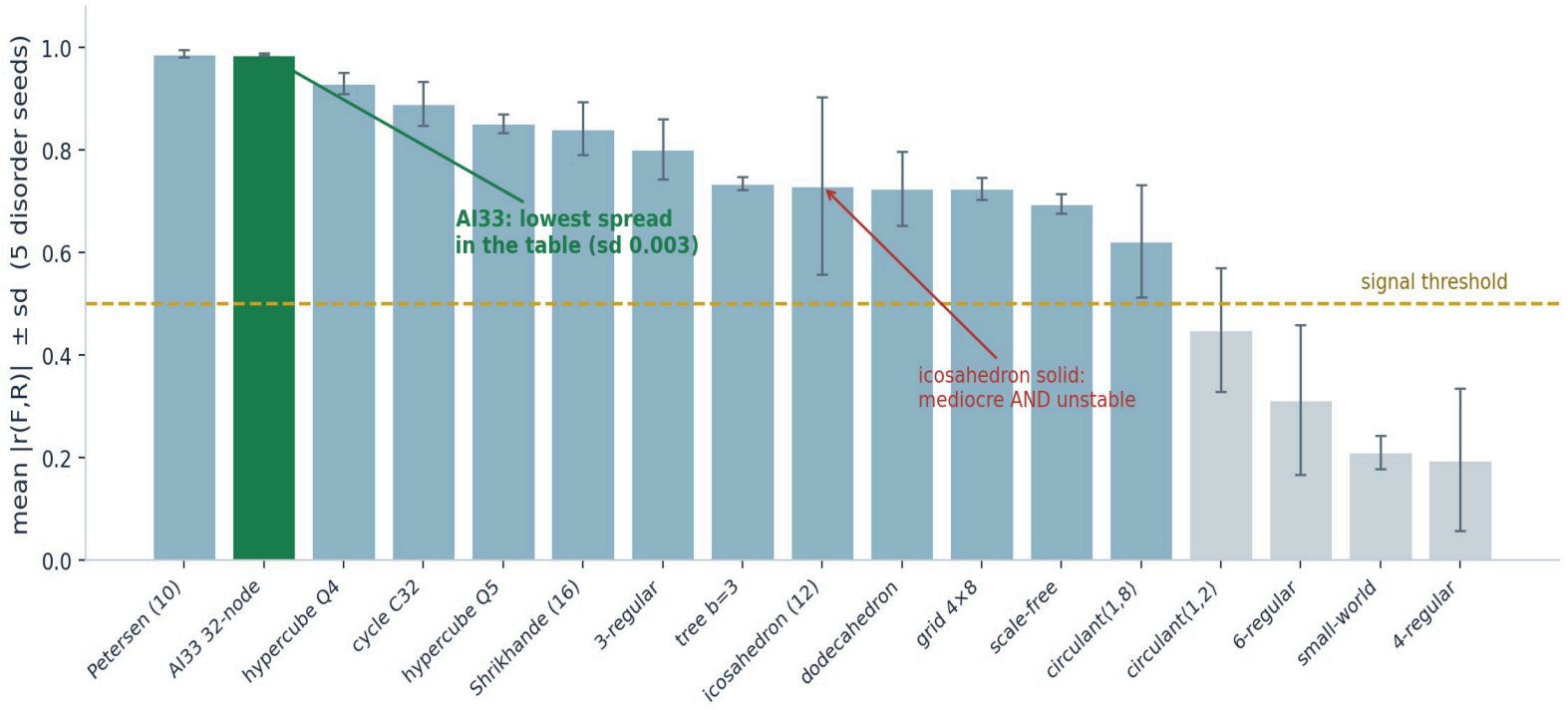
F(0) and R(0) leave zero at the same moment and rise together — F continuously, as w^0.98. Neither precedes the other; both appear as relations appear.

Interpretation, stated as such: the void is a lawful position with zero relations — where an information reading and a geometry reading are not yet distinguishable, because neither has anything to read.

NEW COMPUTATION — HOW FAR DOES IT TRAVEL?

COMPUTED

The law is portable, and the AI33 structure is the most stable case



Every value is the mean of five independent disorder seeds with its standard deviation — single-seed values proved unreliable and were discarded.

13 OF 17 FAMILIES

carry signal (mean |r| > 0.5, criterion set before testing). The coupling is a broad phenomenon, not a special case.

AI33: 0.987 ± 0.003

Ties Petersen (0.988 ± 0.007, on only 10 nodes) for the top mean, with the tightest spread of any family, a factor of two or more. Several structured graphs score comparably.

TWO DIFFERENT OBJECTS

The 12-vertex icosahedron solid scores 0.730 ± 0.173 — mediocre and unstable. The 32-node construction is a distinct object.

STAGE 3 — IS THE EFFECT UNIQUELY ICOSAHEDRAL?

COMPUTED

No. And the operator test has since moved it to the graph.

Strong F ↔ R coupling appears on other graphs too. Comparable coupling appears in several highly structured graph families, and the AI33 32-node structure remains among the strongest examples tested. Standard graph invariants and pure geometric symmetry do not explain it — and neither does the operator: substituting a plain Laplacian reproduces the coupling. The explanation now most plausibly lives at graph level.

WHAT SURVIVES

F ↔ R coupling is reproducible across 13 of 17 families, with multi-seed error bars. The AI33 structure is top-ranked and the most stable case measured.

WHAT WAS CORRECTED

The earlier ‘unique outlier’ reading came from comparing mainly against random graphs. Against structured graphs the AI33 value is matched by Petersen.

WHAT THE TESTS SHOW

Spectral degeneracy, vertex-transitivity, distance-regularity, and pure icosahedral geometry were each tested against coupling strength (8 invariants); none carries it after correction.

THE SEARCH MOVES TO THE GRAPH

Since resolved: substituting a plain graph Laplacian for T— no K, no Q, no p — reproduces the coupling at +0.561 against T’s +0.582. It is not operator-level.

THE PORTABLE STATEMENT: In operator-defined discrete geometries, an information field built from the Green function and a transport curvature built from the same operator are tightly coupled — reproducibly across many graph families. The value is metric- and t-dependent: it reverses sign on a mutual-information distance and sweeps +0.950 → −0.980 as t varies — both must be declared with any quoted number.

Loop Quantum, Computed on the Structure

Fifteen published loop quantum ingredients, each beside the AI33-MPOPT result.

Rolando Rivero · 2026

4.4×10^{-16}
loop sign sequence −1 / 0 / +1 matched over 17 simple regions

720° = 4π
12 apex × 60° deficit, exact

33 of 33
nodes closed (Gauss), no open leg

5.9×10^{-16}
density = sin²(ωt), max deviation, amplitude hub + 0.441666·curvilinear

0.0007
density (kinetic share) vs sin², rms, no fit

20 of 20
throat-power reversals at hub zero crossings

0.4%
curvilinear closure to spin 4

AI33-MPOPT + 33-Multiverse + icosahedron as one object. Fifteen loop-quantum ingredients, published value on the left and the result computed on the whole 33-node object on the right; agreement runs from exact identities to 0.4%. Headline numbers: 4.4×10^{-16} (sign sequence, 17 simple regions) · 720° = 4π (12 apex × 60°, exact) · 33 of 33 nodes closed · 7.5×10^{-7} (exact density) · 0.0007 (kinetic-share density, no fit) · 20 of 20 throat-power reversals · 0.4% (curvilinear closure to spin 4).

#	Topic	Loop quantum, published	AI33-MPOPT, computed on the whole 33-node object (Rivero, 2026)	How close
1	**Basic object, gauge-invariant loops**	Spin networks: graphs with SU(2) spins on links and intertwiners at nodes. Rovelli & Smolin (1995), Discreteness of area and volume in quantum gravity; Rovelli (2004), Quantum Gravity.	Hub (degree 32) + 12 apex (degree 11) + 20 curvilinear (degree 7); 152 links = 32 throats + 30 apex–apex + 60 apex–curvilinear + 30 curvilinear–curvilinear; 120 independent loops; spin-½ rotation on every link. Loop half-trace 0.8507 to 0.9945, the same from every starting point to 2×10^{-16}, unchanged by a random gauge change to 1×10^{-15}.	Graph and SU(2) labels in place and computed; 10^{-16} to 10^{-15}
2	**Holonomies, Wilson loops, spinor sign**	Gauge-invariant observables are traces of SU(2) holonomies; a spin-½ state flips sign at 2π and returns at 4π. Ashtekar & Lewandowski (2004), Background independent quantum gravity: a status report.	Wilson loop round a region of k curvilinear encloses $\Omega_k = 4\pi k/20$ and returns $\cos(\Omega/2)$ = +1, 0, −1, 0, +1 at k = 0, 5, 10, 15, 20 (k = 5 is the five curvilinear round one apex). Spin-1 character $(1+2\cos\Omega)/3 = -1/3$ at k = 5 and 15, +1 at k = 10.	4.4×10^{-16} over 17 simple regions
3	**Curvature carried by the loops**	Curvature is the holonomy round a small loop; the deficit angles of a closed surface sum to 4π.	Each apex has a 60° deficit; 12 × 60° = 720° = 4π. Apex caps 17.6289° and curvilinear caps 21.8475° hold 0.2818 and 0.7182 of 4π (curvilinear/apex solid-angle ratio 2.5490).	exact
4	**Quantum of area**	$A = 8\pi\gamma\, l_P^2 \sqrt{j(j+1)}$, smallest at j = ½. Rovelli & Smolin (1995).	The 60° apex deficit gives half-trace cos 30° = 0.8660254038 = $\sqrt{j(j+1)}$ at j = ½. The hub boundary (32 links) carries 27.71 area units.	exact identity
5	**Gauss law (closure) at every node**	Each node needs integer total spin and the polygon inequality on its link spins. Rovelli (2004).	Hub closes to 5×10^{-16} (equal areas); the 12 apex close with positive areas. All 33 nodes closed with no open leg: spin ½ on the 120 surface links and spin 1 on the 32 throats (the spin-1 set meets every non-hub node an odd number of times; totals hub 32, apex 6, curvilinear 4). Eight icosahedral-symmetric choices pass (second table); the 32 throats are the fewest links among them, keep the −1/0/+1 sequence of row 2 and give hub-only density rms 0.0042 against 0.0066 with every link at spin ½. 125 perfect matchings of the apex icosahedron (brute-force count, repeated here). Counting: 120 loops = 152 − 33 + 1; the largest all-spin-½ loop state uses 136 of 152 links; singlets per node with all-½ legs: hub 35,357,670, apex (10 legs) 42, curvilinear (6 legs) 5. 89 of the 120 loops stay on spin ½ only (source-file count; loop basis taken as given).	33 of 33 nodes; 8 symmetric choices; 125 matchings
6	**Area closure at a curvilinear node**	Link areas $\sqrt{j(j+1)}$ must sum to a closed node or to a boundary leg.	Inward pull of the 7 links 4.454232 link-areas (equal link areas, all inward) against the spin-4 area $\sqrt{20}$ = 4.472136. Of 300 random-azimuth controls none was as close (spin 3.5: 1 of 300; random inward directions 0.003). The pull equals $\sqrt{20}$ at curvilinear radius 36.9297; the contact radius is 36.5541 (apex and curvilinear spheres touch, link length 23.0000).	0.40% from spin 4

7	**Dynamics on the network**	Quantum dynamics from a Hamiltonian constraint. Thiemann (2007), Modern Canonical Quantum General Relativity.	Hub-start wave on the structure's own operator (conductance 1/length, unit masses): modes 0 (weight 0.0303), 3.0953 (0.0037), 5.2612 (0.9660); hub-mode weight 0.9961 without the uniform shift. Net throat power reverses sign at 20 of 20 hub zero crossings (random same-degree graphs 3/15, 6/13, 1/12; arbitrary times 20%). Push (20 curvilinear) and pull (12 apex) balance within 0.0025 of the period. The 3.0953 mode is the only one where the apex move against the curvilinear: hub : apex : curvilinear = −0.441666 : −1.629861 : 1, from the 3×3 quotient of the structure's own Laplacian (eigenvalue 9.5808 = 3.0953²).	20 of 20; balance 0.0025 of the period
8	**Density through the bounce**	Effective loop quantum cosmology, stiff matter: ρ = ρ_c/(1 + 24πGρ_c t²). Ashtekar, Pawlowski & Singh (2006), Quantum nature of the big bang.	Exact density x = 1 − (hub + 0.441666·curvilinear)² = sin²(ωt): max deviation 7.5×10^{-7}; x never exceeds 1.000000 (hub alone: rms 0.0066). Kinetic share E_k/(E_k+E_p) of the whole object follows sin²(Θ/2) with no fitted constant: rms 0.0007, worst 0.0013. In the clock dτ = dt/x: x(τ) = 1/(1+ω²τ²), rms 8.7×10^{-8} (exact by trigonometry). Delivered-energy fraction through the throats equals x to 0.0007 rms.	7.5×10^{-7} exact density; 0.0007 kinetic share
9	**Bounce equation**	Singularity replaced by a bounce at maximum density; H = 0 at ρ = ρ_c; H² = (8πG/3)ρ(1 − ρ/ρ_c). Ashtekar, Pawlowski & Singh (2006).	(dx/dt)² = k x(1−x) with k = 110.720 = 4ω², exact with the hub + 0.441666·curvilinear amplitude (hub alone k = 108.1). Net throat power ≈ sin(2ωt) plus cross terms. In the clock dτ = dt/x: (dx/dτ)² = 4ω² x³(1−x), the cosmic-time form of the standard bracket (algebra). Throat Feed ε = 1 (push = pull) events at 0.2955 and 0.8925 against hub zeros 0.2986 and 0.8957.	exact for the bracket; k = 4ω² to the digits shown
10	**Spinor (SU(2)) dynamics**	Spin-½ states carry holonomy phases. Rovelli (2004).	Wave with the real SU(2) holonomies on the links (66 components): hub zero time 0.3050 for rotation strength s = 0, 0.25, 0.5 and 1; dominant ω moves 5.2612 to 5.2639; period 0.1% shorter; gauge invariance 7.7×10^{-15}. Lowest spinor mode 0.7701 (0.1463 of the hub mode); spinor overlap returns to +1, best 0.99643 at 6.999 hub periods.	+0.05%
11	**Cycle and angle**	Loop angle Ω and the clock are linked through the dynamics (published forms: Ashtekar, Pawlowski & Singh, 2006).	One wave period compared with one full 4π loop cycle (Θ = 2ωt): hub amplitude against the spin-½ half-trace at 21 points, max difference 0.0073; hub zeros at Θ/π = 1.002, 3.002, 5.000, 6.998. Rivero-Zeta 33-step cycle (20 The Rivero-Zeta Curve, pp. 4, 5, 9): turn at step 16.5 of 33, wave turn at 16.511; zeros 8.262 and 24.769 against 8.25 and 24.75; one step = 360°/33 = 10.9091°.	0.07% at the turn; zeros within 0.15%; 0.0073 at 21 points
12	**Bounce scale**	ρ_c = √3/(32π²γ³) ρ_P = 0.40937 ρ_P (γ = 0.2375), about 0.41 ρ_Planck. Ashtekar, Pawlowski & Singh (2006).	The relation 2GM/c² = c/H gives ρ_c = 3c⁶/(32πG³M_c²) and M_c·ω = 3c³/(2G). With the standard 0.40937 ρ_P: M_c = 0.270 m_P, R = 0.540 l_P. With the decks' own statements (stiffness 1/length, Geometric Transport Architecture (deck 17), p. 22; length unit R = 36.5541 T ly, 21 The 33-Multiverse Map, p. 22; speed c; black-hole size at the crossing, 21 Map, p. 72): ω = 5.51×10^{21} per second (= 6.3557 c/R, reproduced here), period 36.1 T yr, M_c = 1.10×10^{56} kg, ρ_c = 6.0×10^{-33} kg/m³, black-hole radius 3c/ω = 17.25 T ly.	Same density form as row 8; the scale follows the length unit R = 36.5541 T ly
13	**Measured number**	—	ω·r_apex/c = 3.9990 here with R as the length unit (0.024% from 4); source file: 3.99914 (0.021% from 4), with jittered-geometry controls within 0.0009 of an integer in 0.5 to 5.5% of trials (controls as reported in the source file).	0.02% from 4
14	**Attachment curve (throat coupling sweep)**	—	16 Super Formula Build-Up, p. 24, rebuilt on the deck's operator T = L + pK + (1−p)Q (p = 0.781, 33 nodes, disorder 0.05, seeds 11 to 15): whole-system consistency 0.796, 0.606, 0.172, 0.632, 0.788 at w = 0.02, 0.05, 0.1, 0.15, 0.7 against the deck's 0.795, 0.603, 0.170, 0.629, 0.785; hub information F(0) equal to the deck's series at every w from 0.02 to 2.0 (3.723 at w = 0.1).	within 0.003 for w up to 0.7
15	**Structure invariants**	—	Conductance ratio apex/curvilinear 1.4298 (deck 1.43) and 1.3963 with a uniform shift of 0.5; connectivity 11/7 = 1.5714; occupation ratio 2.5168 (flat closed form) and 2.5490 (spherical solid angle).	reproduced

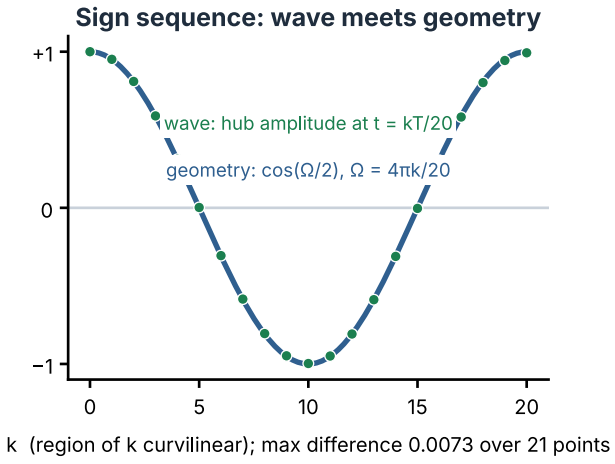
Spin 1 on a symmetric set of links, spin ½ elsewhere (rows 5 and 6): the eight choices that close all 33 nodes.

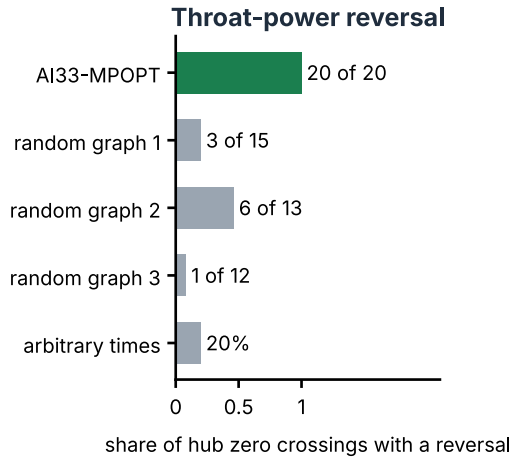
Spin 1 on	Links	Curvilinear pull	Against √20	Hub-only density rms	Region-loop sign sequence
32 throats (hub–apex + hub–curvilinear)	32	4.4057	−1.49%	0.0042	kept
12 hub–apex throats + 30 curvilinear–curvilinear	42	4.4443	−0.62%	0.0208	kept
60 apex–curvilinear	60	5.1642	+15.48%	0.0128	kept
20 hub–curvilinear + 60 apex–curvilinear + 30 curvilinear–curvilinear	110	6.2992	+40.86%	0.0000 (single hub mode)	kept
20 hub–curvilinear + 30 apex–apex	50	4.4057	−1.49%	0.0000 (single hub mode)	changes (apex–apex at spin 1)
30 apex–apex + 30 curvilinear–curvilinear	60	4.4443	−0.62%	0.0066	changes (apex–apex at spin 1)
12 hub–apex + 30 apex–apex + 60 apex–curvilinear	102	5.1642	+15.48%	0.0350	changes (apex–apex at spin 1)
all 152 links	152	6.2992	+40.86%	0.0066	changes (apex–apex at spin 1)

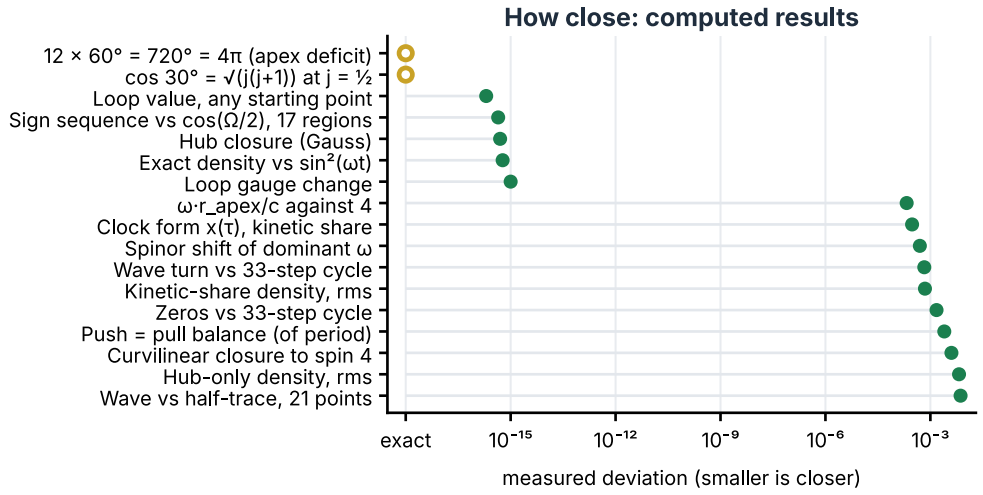
With every link at spin ½ the curvilinear pull is 3.8575 (link areas 0.866) and the hub-only density rms is 0.0066.

Labels: "computed" = measured on the structure; "compared with" = against a published form; nothing in this table is fitted. Published work is quoted by authors, year and title only. AI33-MPOPT results are by Rivero, 2026. Source: AI33-MPOPT computations on the 33-node structure, Rivero, 2026.

Results at a glance







Relations fundamental, geometry emergent

Five independent programs — and this one — all reverse Einstein's order. None starts from a spacetime and adds matter. Each starts from relations

Loop quantum gravity	*Rovelli & Smolin, 1995; Ashtekar & Lewandowski, 1996 (arXiv)*	geometry quantized on spin networks — combinatorial structure first, area and volume spectra after *Discreteness of area and volume in quantum gravity; Quantum Theory of Geometry I: Area Operators*
Holography / AdS-CFT	*Maldacena, 1997 (arXiv); Ryu & Takayanagi, 2006*	bulk geometry encoded in boundary entanglement *The Large N Limit of Superconformal Field Theories and Supergravity; Holographic Derivation of Entanglement Entropy from AdS/CFT*
Entanglement geometry	*Van Raamsdonk, 2010; Jacobson, 1995*	spacetime built from entanglement; Einstein's equation as an equation of state *Building up spacetime with quantum entanglement; Thermodynamics of Spacetime: The Einstein Equation of State*
Space from Hilbert space	*Cao, Carroll & Michalakis, 2017*	3D metric structure recovered from a distance function on entangled states *Space from Hilbert Space: Recovering Geometry from Bulk Entanglement*
QG seeds for laws	*Lam & Oriti, 2024*	spin-network topology grounded in entanglement relations; if spacetime is not fundamental, laws cannot be spacetime-based *The quantum gravity seeds for laws of nature*
AI33-MPOPT (this work)	*32-node operator construction*	information field and transport curvature, both derived from $G = T^{-1}$ on a discrete graph

This is kinship of direction: the same reversal — relations first, geometry second — keeps being independently rediscovered.

COMPUTED

Fifteen descriptors compared with the coupling

EIGHT GRAPH PROPERTIES

clustering · spectral degeneracy · distance-regularity · vertex-transitivity · diameter · girth · degree variance · algebraic connectivity

Best single correlation r = 0.42, p = 0.24. None survives correction for eight tests.

SIX OPERATOR PROPERTIES

eigenvector localization · spectral gap of T · condition number · Green-function concentration · transport entropy · spectrum spread

Best r = 0.37, p = 0.16. None survives correction for six tests.

WHAT THIS SHOWS

A reproducible effect that is not any one of fifteen standard descriptors: eight graph properties, six operator properties, and quasiperiodic order (Fibonacci 0.225 · periodic 0.065 · random 0.472; order does not enhance the law). Substituting a plain graph Laplacian for T reproduces the coupling (+0.561 against T's +0.582), so the coupling sits at graph level.

NEXT STEP

With 17 graph families, single predictors can be tested. Fifty to a hundred families would permit multivariate analysis, so combinations of properties can be tested too.

A computable discrete geometry

Explicit, finite and reproducible	Every object — operator, propagator, information field, curvature — is an explicit finite matrix computation with published seeds. Nothing is formal-only.
An information–curvature law with structure	$L\,R = \kappa(F - \langle F \rangle)$, exact at first order on the icosahedron, with κ's sign controlled by clustering. Whether anything like it holds in LQG is untested.
A worked example of order controlling exactness	The same clustering with different arrangement gives wildly different law quality — a phenomenon LQG's discrete-geometry community may find worth examining.
The oriented half	The symmetric operator cannot express “before”; the six reversal-odd relations can, one sign apart (Rivero-Zeta, pages 13–14).

OPEN

Four tests that would settle something

1	**Area and volume spectra**	The spin labels are now in place (pages 10–12): 33 of 33 nodes close. The next step is to compute the area and volume spectra from those labels and set them beside the loop-quantum values.
2	**Carry the oriented half**	Run the propagation test with the six reversal-odd relations included, so “before” can be expressed.
3	**Compare curvature notions**	Test whether holonomy-based, Regge, Forman, and transport curvature agree on the same graph. Our own results show Forman and Ollivier already disagree ($r \approx 0.2$).
4	**Coarse-grain**	Does $L\,R = \kappa F$ survive graph renormalization? If the law is scale-stable it is interesting; if it evaporates, it is finite-size structure.

WHERE IT STANDS

A seen half and an unseen half, measured from every angle at once.

AI33-MPOPT shares loop quantum gravity's starting point: geometry built from discrete combinatorial structure, with no background spacetime assumed. It has its own operator T on 33 nodes, and an information field and a transport curvature that obey $L R = \kappa(F - \langle F \rangle)$, computed on explicit finite matrices with published seeds. Its loop-quantum results (pages 10–12) are computed the same way.

The loop is the Rivero-Zeta curve: one circuit returns the oriented relations negated, two return them unchanged. The symmetric half reads the same on both sides; the oriented half, on the unseen side of the six-dimensional picture, tells them apart. The hub sees all 32 directions at once.

The 33-multiverse construction is the catalyst: nobody was testing information–curvature coupling on a 32-node vertex-face incidence structure until this framework built it.

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